

Network Fundamentals

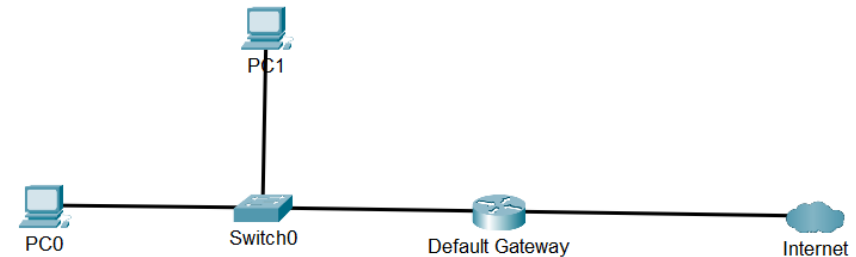
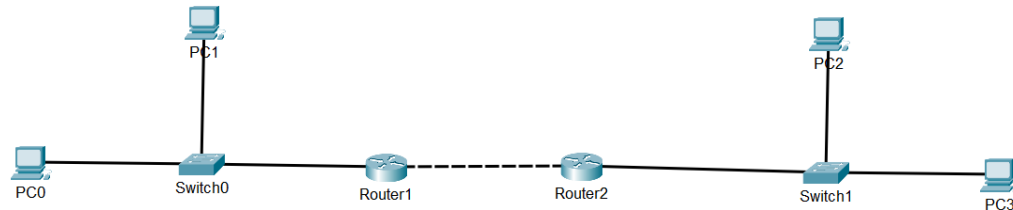
Troubleshooting a network

- When a device is experiencing a problem, it is crucial to know what is happening in the network.
- A lot can go wrong in a network. When trying to find the source of the problem, check each part of the process
- What does work, where does it stop working

Does your computer have an IP address?

Use “ipconfig” or “ip a” to see the IP information

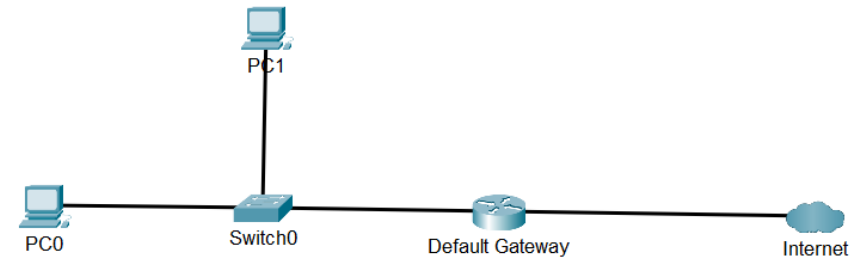
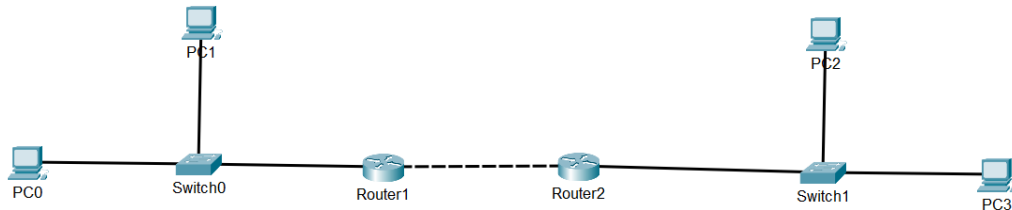
- Do you have a valid IPv4/IPv6 address and subnet mask/prefix length?
- Do you have a valid default gateway and DNS server?
 - Problem with DHCP server
 - Problem with switch in the network
 - Problem with NIC



Network connection on the local network

Use the ping command to check your local network connection

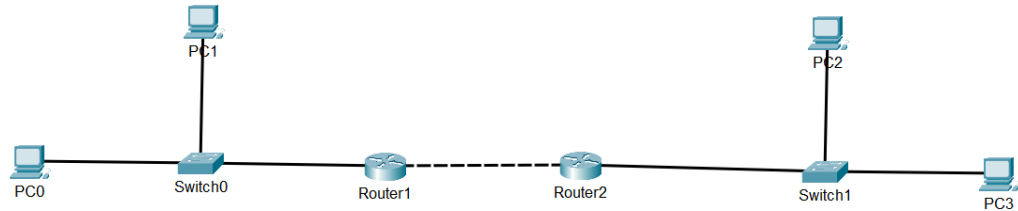
- Can you ping your default gateway
 - Problem with a switch
 - Problem with your default gateway



Network connection on a remote network

Use the ping and tracert command to check the connection to a remote network

- Can you ping the remote network
 - Problem with a router (interface ip / routing table)



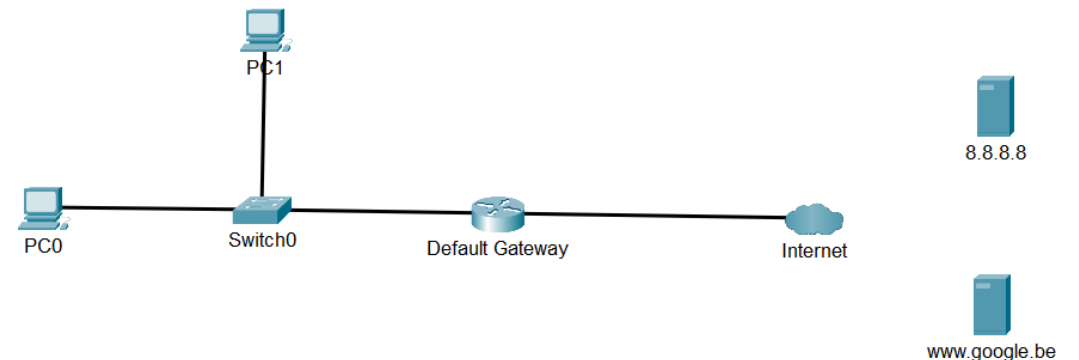
Internet connection

Use the ping command to check your internet connection

- Can you ping 8.8.8.8 (or another well known ip address on the internet)
 - Problem with your router
 - Problem with your internet connection

Use the tracert command to check your internet connection

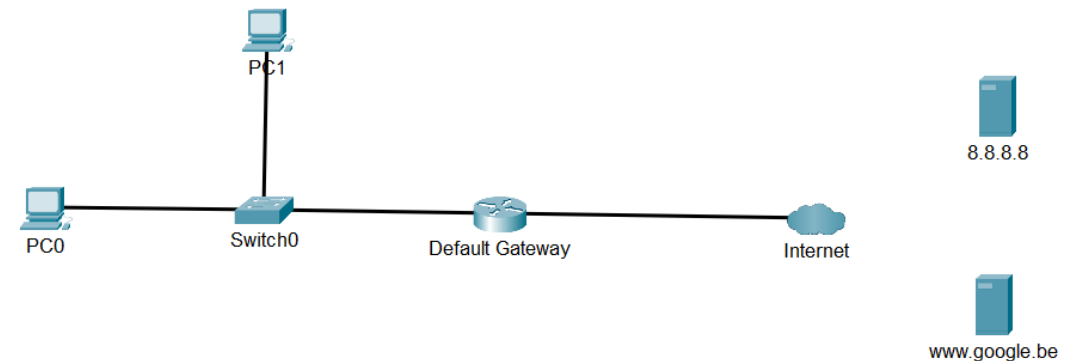
- Can you reach 8.8.8.8
 - You can see if the problem is in the local network or on a remote network



DNS

Use the nslookup command to test dns

- Can you nslookup a public domain name?
 - Problem with your local dns setting
 - Problem with dns server



Application problem

Use ping to try to reach the domain name

- If you can, the network is fine. Check the application
 - Are the other applications reachable (ftp, telnet/ssh, ...)
 - Problem with the application